On the Mechanism Underlying Bursting in the Aplysia Abdominal Ganglion *R*15 Cell

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ABSTRACT

A mathematical model is developed to describe the periodic behavior of the membrane potential of the *Aplysia* *R*15 cell (a pacemaker neuron) in the presence of the drug tetrodotoxin. Using a technique of the qualitative theory of differential equations, this model is shown to be capable of posessing an oscillatory solution representing the bursting electrical behavior. Numerical solutions of the model are shown to predict many of the experimental results reported in the literature on this cell.

INTRODUCTION

Many of the neurons of the *Aplysia* abdominal ganglion display spontaneous activity in one of two common forms: a more or less regular firing pattern known as "beating", or clusters of action potentials separated by relatively long periods of hyperpolarization known as "bursting". An excellent discussion of these phenomena may be found in Ref. 22. Briefly, the bursting behavior has been found to be due to an endogenously arising slow oscillation in the membrane potential, known as the "slow wave", which may be observed when action potentials have been blocked by the drug tetrodotoxin (TTX) [16, 23]. Several hypothesis have been put forth to explain the generation of this slow wave, the two most common being the cyclic variation of an electrogenic sodium-potassium pump [20, 23] and a cyclic variation in the membrane potassium conductance [3, 21].

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Recently, Junge and Stevens [14] have provided convincing evidence that the latter is indeed the case, and that, at least in the R15 cell [11], the slow wave is the result of a cyclic variation in potassium conductance. However, there has not as yet been given any explanation of how such an oscillation in potassium conductance could occur in the R15 cell. In this article, we demonstrate that such a slow wave may be predicted mathematically on the basis of phenomena already observed and documented in molluscan neurons. A qualitative mathematical model of the action of the *Aplysia* R15 cell under the influence of TTX is derived, and this model is shown to predict many of the experimentally observed behavior patterns of the cell.

BACKGROUND

The foundations for the theoretical investigation of neuronal membrane phenomena were laid by Hodgkin and Huxley [13] over twenty years ago. Their model of the behavior of the squid giant axon is by now so well known that familiarity with it will be assumed. Excellent reviews of the subject are found in Refs. 9 and 19. Since the work of Hodgkin and Huxley on the squid, similar models have been developed for many other preparations, among them the myelinated nerve of the toad [10], the Purkinje fiber [8], the snail neuron [12], and the marine molluscan central neuron [5]. All of these models are phenomenological in nature, that is, they do not attempt to explain the actual underlying causes of the membrane behavior, but rather they simulate the observed behavior through hypothetical variables (called activation and inactivation terms) whose behavior is inferred from experimental observation. This philosophy, which is in the spirit of the ancient astronomers who used epicycles to predict the motion of the planets before the theory of celestial mechanics was understood, will be adhered to in this paper.

In a comparison of the behavior of pacemaker (i.e., endogenously active) and nonpacemaker neurons in the *Aplysia* abdominal ganglion, Alving [1] reported that, under a step change in clamping voltage, while the behavior of nonpacemaker cells was qualitatively similar to that of the squid giant axon, the pacemaker cells showed a distinct drop in the maintained outward current under long voltage steps (Fig. 1). These experiments were carried out on somata which had been separated from their ganglion and isolated from contact with their axon, thus ensuring freedom from synaptic interference (invertebrate central neurons in general are synapsed upon on their axons). In addition, Alving reported that the somata were approximately equipotential under voltage clamp, thus justifying the use of ordinary differential equations in a model of their behavior. A similar

drop in outward current was observed in the *Aplysia* R15 cell by Faber and Klee[7].

Connor and Stevens [5], in a quantative analysis of central neurons of nudibranch molluscs, found a similar drop in current, and identified it as being due to a second potassium conductance, independent of the term corresponding to the Hodgkin-Huxley g_K, which displayed inactivation as well as activation. Connor and Stevens also identified, but did not include in their model, a third component of potassium activation with a time constant much longer than any in the model, which they believed was responsible for the adaptation which the cell displayed in the presence of maintained depolarizing current. The object of this paper will be to demonstrate that such a three-component potassium system, coupled with a tonic electrogenic pump current, is sufficient to cause, under certain conditions, the generation of the slow waves associated with bursting cells in the presence of TTX.

The following notation will be used throughout the paper.

V: membrane voltage.

C: membrane capacitance.

V_I: Nernst equilibrium potential of the combined inward-flowing ions (assumed to be Na^+ and Ca^{++}).

V_K: Nernst equilibrium potential of K^+ ion.

X_i: activation term.

Y_i: inactivation term.

$\alpha_i(V)$: activation time constant.

$\beta_i(V)$: inactivation time constant. $i = I, K, A, P, L$

$S_i(V)$: steady state activation [assumed to be an S-shaped function of V; See Fig. 2(a)].

$Z_i(V)$: steady state inactivation [assumed to be a Z-shaped function of V; See Fig. 2(a)].

g_i: maximum ionic conductance.

g_T: inward conductance in the presence of TTX.

t: time.

I_{ep}: tonic electrogenic pump current.

η: normalized voltage [$\eta = (V - V_K)/(V_I - V_K)$].

μ: normalized pump current ($\mu = I_{ep}/(V_I - V_K)$].

γ_i: normalized ionic conductance ($\gamma_i = g_i/C$, $i = I, K, A, P, L, T$).

The subscripts I, K, A, P, L denote respectively the inward, fast outward, slow outward, very slow outward, and leakage current conductances (the first three are denoted I, K, and A by Connor and Stevens). To avoid confusion between the A channel and the activation terms, we have re-

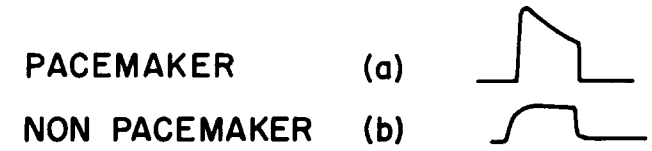


FIG. 1. Comparison of response of voltage clamped pacemaker (a) and nonpacemaker (b) cells to long term step depolarizations. Redrawn from Ref. 1.

placed the symbols of Connor and Stevens for the activation and inactivation process, A_i and B_i, with X_i and Y_i.

The functions $S_i(x)$ and $Z_i(x)$ are defined as having the following properties (see Fig. 2).

$$\begin{array}{lll} & S_i(x) & Z_i(x) \\ \text{(a)} & \lim_{x\to\infty} S(x)=1, & \lim_{x\to\infty} Z_i(x)=0, \\ & \lim_{x\to-\infty} S_i(x)=0, & \lim_{x\to-\infty} Z_i(x)=1, \\ \text{(b)} & \dfrac{dS_i}{dx}\geqslant 0, & \dfrac{dZ_i}{dx}\leqslant 0, \end{array}$$

DEVELOPMENT OF THE MODEL

The model of Connor and Stevens, in the notation of the previous section, is

$$\dot{V}=-\frac{1}{C}[g_{\mathrm{I}}X_{\mathrm{I}}^3Y_{\mathrm{I}}(V-V_{\mathrm{I}})+g_{\mathrm{K}}X_{\mathrm{K}}^3(V-V_{\mathrm{K}}) + g_{\mathrm{A}}X_{\mathrm{A}}^4Y_{\mathrm{A}}(V-V_{\mathrm{K}})+g_{\mathrm{L}}(V-V_{\mathrm{L}})], \tag{1a}$$

$$\dot{X}_i=\frac{1}{\alpha_i(V)}[S_i(V)-X_i], \qquad i=\mathrm{I},\mathrm{K},\mathrm{A}, \tag{1b}$$

$$\dot{Y}_i=\frac{1}{\beta_i(V)}[Z_i(V)-Y_i]$$

where the dot denotes differentiation with respect to time. There are two modifications which have been made to the Connor-Stevens equations. First, the two-component potassium K activation, which gave a slightly improved curve fit, has been eliminated, and second, the Nernst equilibrium potentials of the A and K potassium components, which were slightly different, have been set equal. If we now add a tonic hyperpolarizing pump

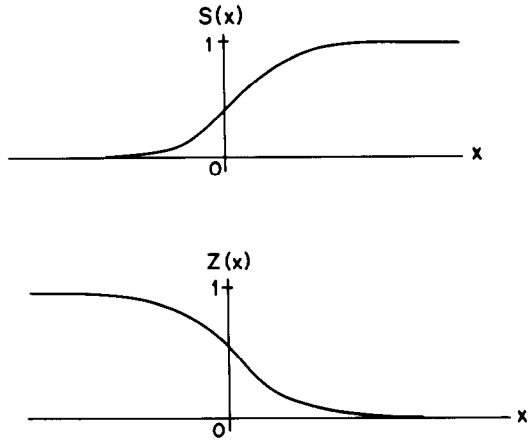


FIG. 2 Examples of functions $S_i(x)$ and $Z_i(x)$.

current term and a very slow potassium activation term, Eq. (1a) becomes

$$\dot{V} = I_p - \frac{1}{C}\left[g_I X_I^3 Y_I (V - V_I) + (g_K X_K^3 + g_A X_A^4 Y_A + g_p X_p)(V - V_K) + g_L (V - V_L)\right], \quad (2a)$$

and i also takes on the value p in Eq. (1b).

Let us now suppose that TTX is added to the bath surrounding our hypothetical cell. We will make the assumption that the effect of this drug is to fix the inward conductance $g_I X_I^3 Y_I$ at a constant value g_T. That is, TTX is assumed to abolish the transient Na^+ current. In the presence of TTX, the R15 membrane is observed to oscillate in a manner such as that shown in Fig. 3, with the membrane oscillating in slow waves between -30 mV and -70mV. Since the fast component of the potassium conductance is less than 7 percent of its maximum value when the membrane voltage is below -20 mV [5, Fig. 5], we assume that this component is never activated, and that, in the presence of TTX,

$$g_K X_K^3 \equiv 0.$$

Since g_L is two orders of magnitude less than the other conductances, we shall neglect it, and since we have no sure information about $\alpha_p(V)$ except that it is large, we assume it to be a constant, α_p, with

$$\alpha_A(V), \beta_A(V) \ll \alpha_p$$



FIG. 3. Membrane slow wave oscillation in the presence of TTX. Redrawn from Ref. 16.

for all values of V. Now let

$$\eta = \frac{V - V_K}{V_I - V_K}, \qquad \mu = -\frac{I_{ep}}{V_I - V_K}, \qquad \gamma_i = \frac{g_i}{C}, \quad i = T, A, P.$$

Assuming the presence of TTX and neglecting g_L, we have

$$\begin{aligned}
\dot{\eta} &= \gamma_T[1-\eta] - [\gamma_A X_A^4 Y_A + \gamma_p X_p]\eta - \mu, \\
\alpha_{A\eta}(\eta)\dot{X}_A &= [S_{A\eta}(\eta) - X_A], \\
\beta_{A\eta}(\eta)\dot{Y}_A &= [Z_{A\eta}(\eta) - Y_A], \\
\dot{X}_p &= \frac{1}{\alpha_p}[S_{p\eta}(\eta) - X_p],
\end{aligned} \tag{3}$$

where the η subscripts indicate that the functions are now expressed in terms of η instead of V.

We shall now demonstrate that under certain conditions the system (3) possesses a solution whose form is the same as that of the slow wave. Rather than studying this fourth order system directly, we shall demonstrate that the solution of the system approaches asymptotically the solution of a second order system and then study the behavior of that second order system. This technique is similar to that used by Krinskii and Kokoz [15] in a study of the Hodgkin-Huxley equations. It involves the reduction of a multiple time constant n-dimensional dynamical system to a system of m differential and n-m algebraic equations. In particular, if $m=2$, then the phase plane methods of analysis become applicable. Fundamental to this technique is a theorem of Tikhonov, stated in the appendix, which provides a simple criterion for determining whether the solution of a multiple time constant system approaches that of a system of smaller dimension asymptotically as the small time constants approach zero.

We will show that the system (3) satisfies the criterion of Tikhonov's theorem. In the notation of the appendix, we wish to examine the stability of the root of the system $\mathbf{v} = \psi(t, \mathbf{u}) = \{S_{A\eta}(\eta), Z_{A\eta}(\eta)\}$, letting the region D be the entire (η, X_p, X_A, Y_A) phase space. We have

$$F(t, u_1, v_j) = -(S_{A\eta} - X_A)^2 - (Z_{A\eta} - Y_A)^2 < 0 \qquad \text{for } S_{A\eta} \neq X_A,\ Z_{A\eta} \neq Y_A.$$

Thus $F(t,u_i,v_j)$ is uniformly negative, and for arbitrary ϵ_o, the $\dot{x}_A = \dot{Y}_A = 0$ null surface is a stable manifold whose region of influence is the whole space. Therefore as $\alpha_{AH}(\eta)\to 0$, $\beta_{A\eta}(\eta)\to 0$, the solutions of the system (3) approach those of the system

$$\dot{\eta} = \gamma_T[1-\eta] - [\gamma_A S(\eta) Z(\eta) + \gamma_p X_p]\eta - \mu,$$

$$\dot{X}_p = \frac{1}{\alpha_p}[S_p(\eta) - X_p], \tag{4}$$

$$X_A = S(\eta)^{1/4}, \qquad Y_A = Z(\eta),$$

where

$$S(\eta) = [S_{A\eta}(\eta)]^4, \qquad Z(\eta) = Z_{A\eta}(\eta).$$

Remark. Note that the time constant for η is also much smaller than α_p. However, as we shall see, the interesting case is specifically that in which the $\dot{\eta} = 0$ manifold does *not* satisfy Tikhonov's theorem.

We shall now derive the conditions sufficient for the system (4) to posess a unique stable limit cycle in the (η, X_p) phase plane. Since, as mentioned above, the time constant α_p is much greater than that of η, we may therefore use the theory of discontinuous oscillations [2,27]. If we write (4) as

$$\dot{X}_p = f(\eta, X_p),$$

$$\dot{\eta} = g(\eta, X_p). \tag{5}$$

Then the solution of the system (5) will follow that of the system

$$g(\eta, X_p) = 0,$$

$$\dot{X}_p = f(\eta, X_p), \tag{6}$$

except in those regions of phase space where $\partial g/\partial \eta > 0$. In order for the system (5) to have a limit cycle (Fig. 4), there must exist an interval on the nullcline $g(\eta, X_p) = 0$ where $\partial g/\partial \eta > 0$, and the nullcline $f(\eta, X_p) = 0$ must intersect $g(\eta, X_p) = 0$ in this interval (see Ref. 2, p. 659).

Making use of the implicit function theorem, we obtain an equation for the $\dot{\eta} = 0$ nullcline,

$$X_p = \frac{1}{\gamma_f}\left[\frac{\gamma_T - \mu}{\eta} - \gamma_T - \gamma_A S(\eta) Z(\eta)\right] \triangleq G(\eta),$$

and the condition that $\partial g/\partial \eta > 0$ is equivalent to $G'(\eta) > 0$, that is, the

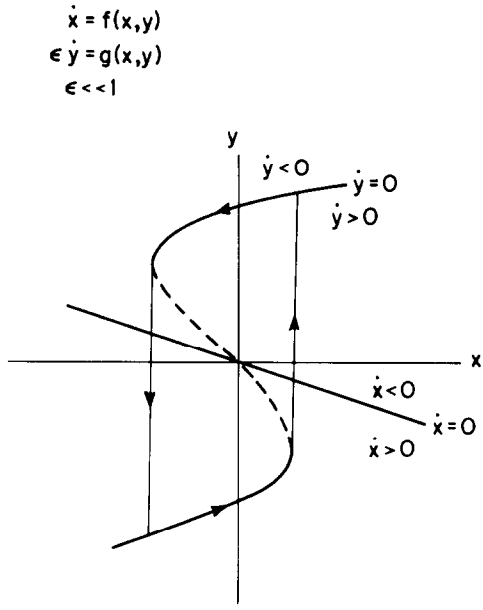


FIG. 4. Limit cycle in a discontinuous relaxation oscillator. The dashed portion of the $\dot{y}=0$ nullcline is the region in which $\partial g/\partial y > 0$.

graph of $G(\eta)$ must have a region with a positive slope. We have

$$G'(\eta)=\frac{1}{\gamma_P}\left[\frac{\mu-\gamma_T}{\eta^2}-\gamma_A[S'(\eta)Z(\eta)+S(\eta)Z'(\eta)]\right].$$

Since by definition (see Fig. 2) $S'(\eta)\geqslant 0$ and $Z'(\eta)\leqslant 0$, we see that $G'(\eta)$ is uniformly negative for $0\leqslant \mu \leqslant \gamma_T$ except where

$$Z'(\eta)S(\eta)<\frac{\mu-\gamma_T}{\gamma_A\eta^2}-S'(\eta)Z(\eta). \tag{7}$$

Let us suppose that the condition (7) is satisfied in some region. We may obtain a qualitative picture of the $G(\eta)=0$ nullcline in the (η, X_p) phase space by noting that as $\eta\to\infty$, $G(\eta)\to-\gamma_T/\gamma_p$, since $S(\eta)\to 0$, $\eta\to 0$, $G(\eta)\to\infty$. Let $S(\eta)$ and $Z(\eta)$ be such that the condition (7) is satisfied on a finite connected, open interval. Then the $\dot{\mu}=0$ nullcline will have the appearance of Fig. 5(a).

Similarly, the $\dot{X}_p=0$ nullcline is given by the equation

$$X_p=S_p(\eta),$$

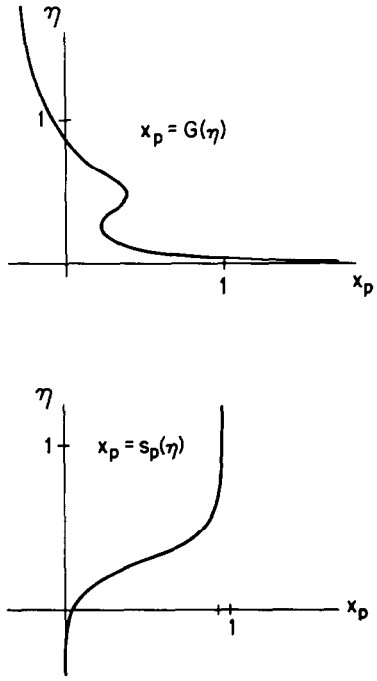


FIG. 5. (a) $\dot{\eta}=0$ nullcline for $Z'(\eta)S(\eta)<\mu-\gamma_T/\gamma_A\eta^2-S'(\eta)Z(\eta)$ on a finite, open, connected interval. (b) Typical $X_p=0$ nullcline.

which has the form of Fig. 5(b). In order for the system (5) to have a unique stable unit cycle, we must be able to combine Fig. 5(a) and (b) to get Fig. 6. If this is to be the case, the two nullclines must have a unique intersection and this intersection must lie in the region of positive slope $G'(\eta)$. Therefore, if the condition (7) is satisfied if and only if $\eta_1<\eta<\eta_2$, then a sufficient condition for the system (4) to have a unique stable limit cycle is that

$$\text{(a)} \qquad G(\eta_*)=S_p(\eta_*) \text{ for some } \eta_*\epsilon(\eta_1,\eta_2)$$

and

$$\text{(b)} \qquad S_p'(\eta)>G'(\eta) \text{ for all } \eta\epsilon(\eta_1,\eta_2).$$

If these conditions are satisfied, then the behavior of the system (4) will be identical with that of the *Aplysia* R15 cell in TTX. In the next section we shall construct a quantitative example of the system and compare its solutions with those obtained experimentally.

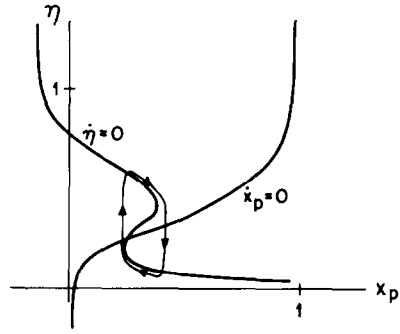


FIG. 6. Relaxation oscillation of $\eta - X_p$ system.

DEVELOPMENT OF AN EXEMPLARY SYSTEM

In this section we shall develop a simple system which satisfies the conditions put forth in the previous section, and compare the behavior of numerical solutions of this system with the observed behavior of the R15 cell in the presence of TTX. The system will be expressed in terms of the normalized parameters of the last section. The time constants were chosen for purposes of computational efficiency, so that the parameter t is nondimensional. In addition, the time constant of the X_A process was increased in order to avoid unduly small step sizes in the numerical integration. Recalling the results of the last section, for sufficiently small magnitude of this time constant, its value does not affect the properties of the solution. For our purposes, the simplest approach is to construct suitable piecewise linear functions for $S(\eta)$, $Z(\eta)$, and $S_(\eta)$. These functions are shown in Fig. 7. The multiplicative terms were all given constant values, these values are

$$\gamma_I = 0.6, \qquad \gamma_A = 50, \qquad \gamma_p = 5,$$

$$\alpha_{A\eta} = 0.1, \qquad \beta_{A\eta} = 0.2, \qquad \alpha_p = 10.$$

With these parameters, the two-dimensional system (4) has the phase plane nullclines shown in Fig. 8 for various values of μ. It can be seen that for $\mu = 0$ and $\mu = 0.05$, the system has a stable equilibrium point; for $\mu = .10$, the equilibrium is unstable, and for $\mu = 0.15$ and 0.20 it is again stable. Thus, for "currents" μ in the neighborhood of 0.10, the system (9) will have a stable limit cycle solution.

Figure 9 shows a plot of η vs t for $\mu = 0.10$ for the systems (3) and (4). System (3) leads system (4) due to the lack of delay in the motion of X_A and Y_A. Figure 10 shows a comparison of the solutions of the system (3) for various values of the "current" term μ, and, for comparison, a series of

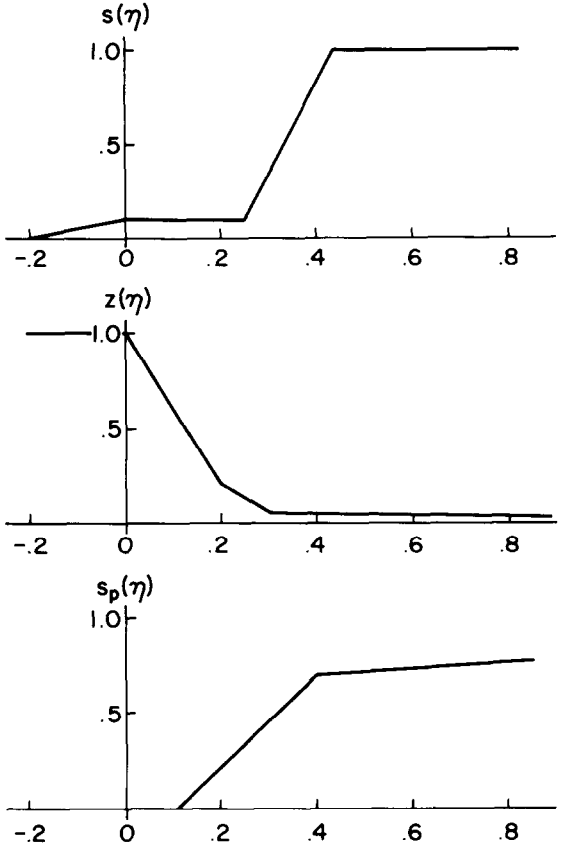


FIG. 7. Piecewise linear S and Z functions used in the model.

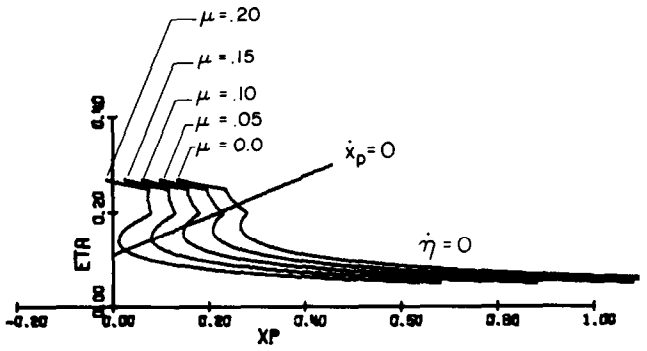


FIG. 8. Computer generated nullclines of the system (4) for various values of μ.

oscillograph tracings of the membrane potential of an R15 cell in TTX with current injection of varying levels, as reported by Mathieu and Roberge [16]. It may be seen that the amplitude and period of the oscillations of both systems increase with increasing inward current.

The tendency of the period of the system (3) to increase with increasing values of μ can be explained by reference to the system (4). This system is topologically equivalent to the Van Der Pol oscillator with a nonzero right hand side, i.e.,

$$\ddot{x} - \epsilon[1 - x^2]\dot{x} + x = h, \qquad h = \text{const.}$$

The dependence of the period of the system on the value of h is a generic property of relaxation ocsillators [9,21,26]. The amplitude of the solution of the Van Der Pol relaxation oscillator is not, however, a function of h. Indeed, the tendency of the amplitude of the solutions of the systems (3) and (4) to increase with increasing μ is due to the particular nature of these systems. Referring to the system (4), the equation of the $\dot{\eta} = 0$ nullcline is

$$X_p = \frac{1}{\gamma_p}\left[\frac{\gamma_T - \mu}{\eta} - \gamma_I - \gamma_A S(\eta) Z(\eta)\right] = G(\eta, \mu),$$

and, as μ is increased from zero, $\mu/\gamma_p\eta$ is subtracted from $G(\eta, 0)$. The subtraction of a quantity inversely proportional to η causes the curve to fold under itself, resulting in a larger amplitude oscillation, as shown in Fig. 8. Furthermore, this amplitude increase will be due to a decrease in the minimum value of the oscillation, while the maximum value stays roughly constant. This is precisely the case for the oscillations of the R15 cell under various injected currents, reported by Mathieu and Roberge.

Mathieu and Roberge also report that for sufficiently large depolarizing injected current, the slow wave is abolished and the membrane potential is maintained at a steady state near the maximum value of the slow wave; similarly, for a sufficientlylarge hyperpolarizing current, the membrane potential is maintained at a steady state near the slow wave's minimum value. This behavior is predicted by the model, and is a consequence of the movement of the singular point into a region where it is stable, either above the region of instability (Fig. 8, $\mu > 0.05$) or below it (Fig. 8, $\mu < 0.15$).

DISCUSSION

The model presented in this paper, which was derived from known properties of molluscan pacemaker neurons, is able to predict many of the observed behavior patterns of the slow wave of the *Aplysia* R15 cell. An examination of the equations (3) shows that this system has much in

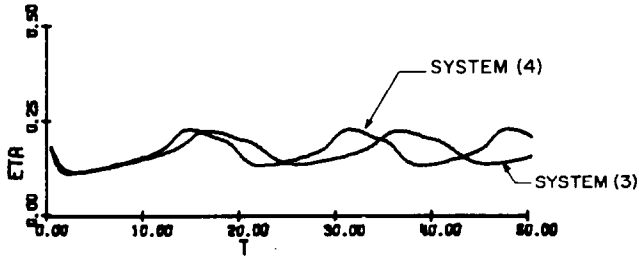
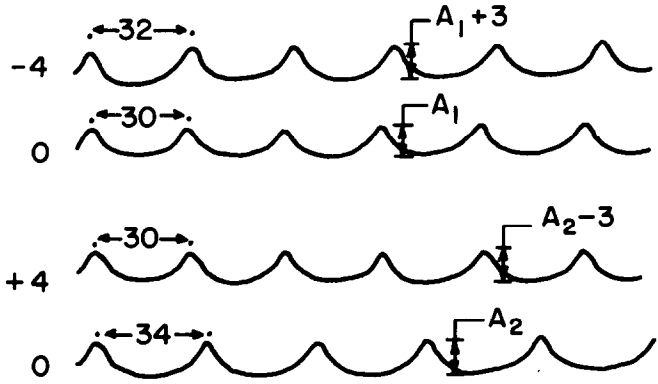
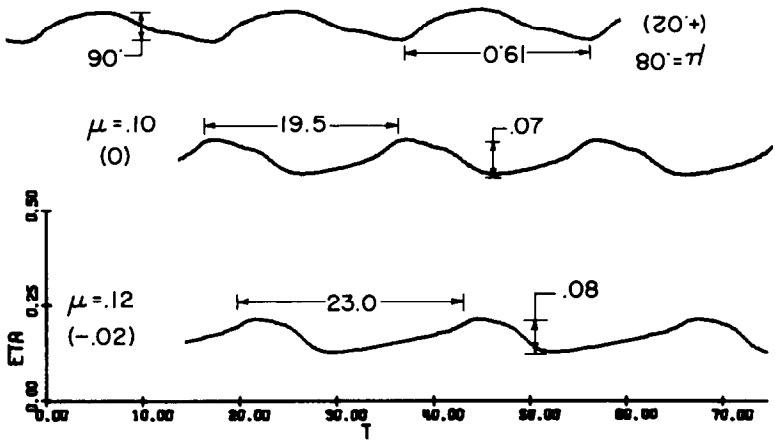


FIG. 9. Numerical solution of the systems (3) and (4). $\mu = 0.10$.



(a)



(b)

FIG. 10. (a) Oscillations in $R15$ cell under various currents. Redrawn from Ref. 16. Figures to the left of the curve denote current value. (b) Numerical solution of the system (3) for various values of "current" μ. "Zero external current" is taken as $\mu = 0.10$.

common with Fitzhugh's modification of the Hodgkin-Huxley equation to simulate the effect of TEA [8], but that the roles of potassium and sodium, and of activation and inactivation, have been interchanged (the slow term is a potassium activation instead of a sodium inactivation). Thus, the long hyperpolarization of this model is dynamically equivalent to the plateau of Fitzhugh's model. The lack of a "spike", as seen in Fitzhugh's model, is due to the fact that while the roles of activation and activation have been interchanged, the relative magnitudes of their time constants have not. Using this interpretation, the condition (7) may be interpreted as saying that the regenerative term (which has the unfortunate name inactivation) must be growing before the degenerative term (activation) becomes small enough to wipe out its effect. Thus the biological implication of condition (7) is that the A channel must retain a residual level of activation in the region where its inactivation term is changing rapidly. This residual activation is present in the model in the region $0 \leqslant \eta \leqslant 0.2$; in the Connor-Stevens model this residual activation is not present, and hence the Connor-Stevens model does not display bursting.

Other phenomena observed in the *Aplysia* R15 cell may also be interpreted in terms of this model. We shall discuss three of them here: the effect of ouabain, the phenomena of "resetting", and the presence of two stable states in some cells. Ouabain is known to block the electrogenic sodium-potassium pump. Its effect on the output of R15 in TTX (also Ca^{++}-free in this experiment) is shown in Fig. 11(a). When ouabain is added, the current, represented by μ, is reduced form its normal value, $\mu = 0.10$, to zero. The singular point of the system thus becomes stable, and the phase point moves from the limit cycle to this stable singular point. The phenomenon of "resetting" [22] may also be explained by reference to the phase plane of the system (9). The injection of hyperpolarizing current during a burst resets the appearance of the next burst (see Fig. 12 and Ref. 22); as sketched in Fig. 12, the effect of a hyperpolarizing current is to move the phase point off the limit cycle. If moved a sufficient distance, it jumps across to a new position and is thus "reset". Finally, it is known [16,23] that during certain periods in the circadian rhythm, the cell does not burst, and that during these periods, if placed in TTX, the cell will jump between two stable steady states. This may be explained as a reduction in the slope of $S_p(\eta)$ sufficient to cause three intersections of the nullclines (Fig. 13), with one unstable and two stable singular points.

Recent published evidence indicates that other mechanisms may also play a role in the behavior of bursting cells. For instance, Wachtel and Wilson [25] have reported evidence of a voltage-dependent electrogenic pump in left half ganglion cells in the *Aplysia* abdominal ganglion. Meech [17] has observed that increases in the internal concentration of Ca^{++} in

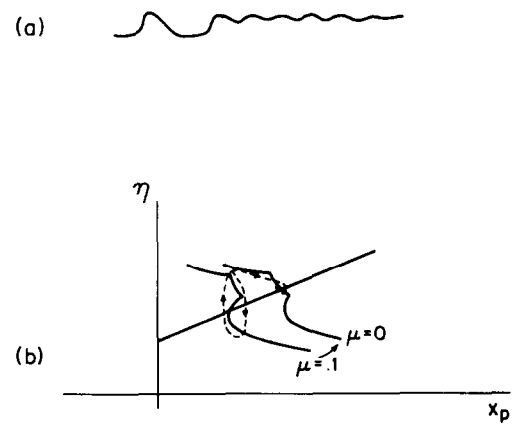


FIG. 11. (a) The effect of the addition of ouabain to bathing liquid already containing TTX (and also Ca^{++}-free). Redrawn from Ref. 14. (b) Phaseplane diagram of movement of $\dot{\eta}=0$ nullcline from $\mu=0.10$ to $\mu=0$, and resulting stabilization of η near peak of oscillation.

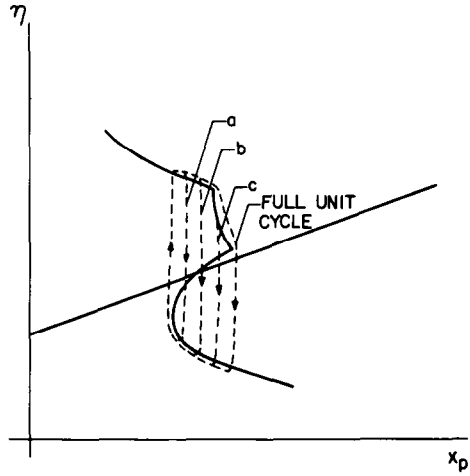


FIG. 12. Phase plane explanation of the resetting of a burst by artificial hyperpolarization of the previous burst. The full unit cycle represents the trajectory of the limit cycle. The dashed lines a,b,c represent limit cycles artificially reset by hyperpolarizing current.

snail neurons increases the membrane permeability to K^+. However, Junge and Stevens obtained slow waves in R15 cells bathed in Ca^{++}-free medium. Eaton [6] has reported the presence of K^+ accumulation in addition to slow K^+ activation in R15. The voltage-dependent pump current could act in

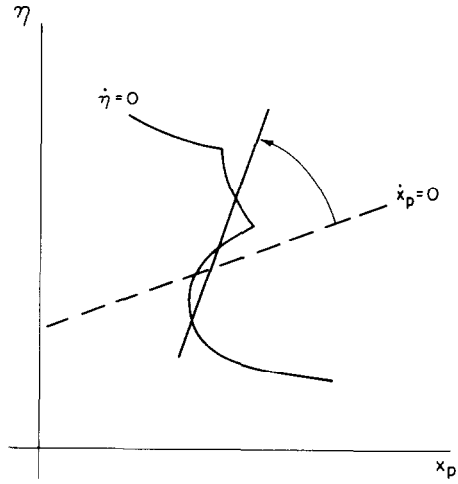


FIG. 13. Reduction in slope of $S_p(\eta)$ resulting in two stable equilibrium points.

cooperation with the K^+ inactivation to produce the regenerative outward current which is fundamental to the generation of the hyperpolarizing component of the slow wave. Similarly, the accumulation of external potassium ions could act in concert with the slow reduction in potassium activation X_p to produce the slow wave's depolarizing component.

APPENDIX: TIKHONOV'S THEOREM

Fundamental to the development described in this paper is the fact that, under certain conditions, the motion of an n-dimensional real dynamical system in its phase space (here assumed to be R^n), is confined to a region of that space which lies near an m-dimensional manifold of that space, where $m < n$. This is true if m of the components of the system have time constants whose magnitude is much less than those of the other n-m components, and if the manifold consisting of the Cartesian product of the nullclines of these m components is stable. Zeeman [27] presents an excellent intuitive discussion of this behavior.

These notions have been made precise in a theorem of Tikhonov [24] which has been translated into English for the convenience of the reader and included here. The proof requires extensive development and has been omitted. Let there be given a system in a region D of its phase space:

$$\frac{du_i}{dt} = f_i(t, u_i, v_j), \qquad i = 1, \ldots, p,$$

$$\epsilon_j \frac{dv_j}{dt} = F_j(t, u_i, v_j), \qquad j = 1, \ldots, m, \tag{A1}$$

with initial conditions $\epsilon_i(t_0)=u_{i0}$, $v_j(t_0)=v_{j0}$ where the ϵ_j may be functions of the other variables, and where the f_i and F_j are bounded and differentiable. Assume that each ϵ_j is less than or equal to $\epsilon_j(\epsilon)$, for some parameter ϵ, and that $\lim_{\epsilon\to 0}\epsilon_j(\epsilon)=0$. Let the root of the system, $F_j(t,u_i v_j)$ be given by

$$vj=\psi_j(t,u_i), \qquad i=1,\ldots,p.$$

Then this root is stable if there exists an r_0 such that the function

$$F(t,u_i,v_j) \triangleq \sum_{j=1}^{m}\left[v_j-\psi_j(t,u_i)\right]F_j(t,u_i,v_j)$$

is negative for all values of t,u,v for which

$$v_j\neq\psi_j(t,u_i) \quad \text{and} \quad r=\sqrt{\sum_{j=1}^{m}\left[v_j-\psi_j(t,u_i)\right]^2} < r_0.$$

Define the region of influence of the stable root $v_j=\psi_j(t,u_i)$, $j=1,\ldots,m$ to be those initial points (t_0,u_{i0},v_{j0}) such that

$$\operatorname{sgn}F(t_0,u_{i0},v_j)=\operatorname{sgn}F(t_0,u_{i0},v_{j0})$$

for all v_j in the region $v_{j0}\leqslant v_j<\psi(t_0,u_{i0}),i=1,\ldots,p,\ j=1,\ldots,m$, i.e.,

$$F_j(t_0,u_{i0},v_j)>0 \quad \text{for} \quad v_{j0}\leqslant v_j<\psi(t_0,u_{i0}) \quad \text{if} \quad v_{j0}<\psi_j(t_0,u_{i0}),$$

$$F_j(t_0,u_{i0},v_j)<0 \quad \text{for} \quad v_{j0}\geqslant v_j>\psi(t_0,u_{i0}) \quad \text{if} \quad v_{j0}>\psi_j(t_0,u_{i0}).$$

THEOREM (TIKHONOV).

If the initial point (t,u_{i0},v_{j0}) is contained in the region of influence of the stable root $v_j=\psi_j(t,u_i)$ of the system (A1), *then as $\epsilon\to 0$ the integral curve $\{(t,u_i(t,\epsilon),v_j(t,\epsilon))\}$ corresponding to the initial point (t_0,u_{i0},v_{j0}) approaches the limit $\{t,\bar{u}_i(t),\bar{v}_j(t)\}$, the solution of the degenerate system*

$$\frac{d\bar{u}_i}{dt}=f_i(t,\bar{u}_i,\psi_j(t,\bar{u}_i)), \qquad j=1,\ldots,m,$$

$$\bar{v}_j=\psi_j(t,\bar{u}_2(t)),$$

$$\bar{u}_i(t_0)=u_{i0},$$

uniformly in the region $t\geqslant t_l>t_0$.

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